

ActInf Textbook Group ~ Cohort 3 ~ Meeting 10 (Chapter 5, part 1)

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Hello, thanks everyone for joining. It's March 29th, 2023. We're in Cohort 3, having our first discussion on Chapter 5. So, before we go into any questions or look at the text, does anyone want to just provide any general reflection or thought on Chapter 5 or just the first half of the book up until now? Just unmute and go for it if you want to.

Okay, let's look to Chapter 5.

And then I'm sure people have many thoughts as we get going.

Any general thoughts on 5 or on this opening quotation about plants and animals?

Well, there are plants that move and there does seem to be some thought that there are interconnections between plants, that they have something that, while it is not an animal central nervous system. There are patterns within the architecture that have similar functions and it's the timeline over which they operate as much longer.

I very much agree with that. Here's my PhD advisor, Deborah, and a paper from 1989 describing the plant behavior idea. And then, I believe a more closer to our area is this 2017 paper on plant predictive processing with Friston. So, agreed. This quote, it's a little bit of like timescale dependent. It's just kind of like a... It's a fun first pass, but isn't that interesting that it's like... It's actually not so stark. Or the plants that do respond, like the Venus flycatchers and everything. But here in this chapter, we're going to be focused on the mammalian nervous system and body, not the morphology of the nervous system or morphological development.

Okay. Anyone, please just raise your hand or write something. Otherwise, let's just go through chapter five and just look at some potentially interesting pieces.

This is an interesting concept that not every variable depends on every other variable.

And that's sparsity. And so, there's sparsity in the brain. There are many connections, but of the space of the possible, it's quite sparse. And similarly, in statistical models, the sparser they are, the easier they are to fit. Doesn't mean that it's going to be better. And a lot of modeling is about finding the trade-off point between like model complexity and model accuracy. Because if you enable every edge, it's going to be a really laborious calculation. But if you restrict edges that are actually important, you might have a very rapidly fit model. However, it might not reflect some of the important true causal relationships amongst variables. So, that comes up in different ways.

Let us take a step back from the technical material of chapter four, everyone, breathe a sigh, and turn our attention to the process theories accompanying active inference. Here, they point to the difference between a principle, free energy principle, which is that systems are described by their dynamics on free energy landscapes, and that persistence is associated with reduction of free energy, to the extent that agency is exerted, and a process theory about how any given system may actually implement those kinds of dynamics. So, a principle cannot be falsified. And this is at the root of a lot of questions and commentary around like the falsification

capacity for the free energy principle. And the simplest thing, as far as I've seen, is neither can a linear regression be falsified. It's just not within the space of being confirmed or rejected or falsified by anything at all. It's just a proposed model framework as a principle. Now, any given linear regression, which is a specific hypothesis about how that principle of linear regression or the principle of the L2 norm might be implemented, that given linear model might be adequate or not. More adequate or totally inadequate or however, but it is more subject and answerable to empirical data. So, free energy principle is not really in the space or the game of falsification or not, nor does it even engage with empirical data directly. Whereas, proposed and manifest models are directly addressing empirical data and empirical data speak to their validity. So, none of the above chapters that we've read could be falsified by any observation in the world, or they're just kind of theoretical. Whereas, the models that are going to start to be shown in chapter five are going to be structural and computational hypotheses about models of the nervous system, and thus, they're more addressable and answerable to real data. Jonathan? Would it be reasonable to say that it may or may not be true that the free energy principle is applicable to certain real world systems, but not that the principle itself is wrong, just that it's not applicable?

Jonathan?

What does anyone want to add on that?

Ali, go for it.

Actually, I came across a kind of situation that kind of contradicts some of the claims of active inference or FEP about their scope of applicability recently. So, I'm not sure if you're familiar with Stefan Thurner's recent work, but basically, he claims that complex systems include a set of systems that are only non-Markovian and non-ergodic systems. But on the other hand, we have this notion, we have this requirement for any system that is to be modeled in active inference framework to be ergodic and Markovian systems. Although in some recent literature, they have dropped this requirement for the system to be ergodic. But again, I believe a kind of local ergodicity needs to be present in order for the system to be ergodic. So, as far as I know, the scope of applicability is at least so far not strictly defined or let's say it's not something that can be

restricted in a definitional way, but there are degrees of applicability. So, for instance,

in Dalton's, Dalton's, Dalton's active adeval's weak markup blank, weak blankets paper, he has this notion of degrees of blanketedness that helps us to distinguish between the systems that have effective markup blankets and

the otherwise linear systems that doesn't contain any kind of effective blanketedness. So, briefly, he claims that

for higher dimensional and complex systems, we can apply these Bayesian mechanical framework, but for linear systems, although theoretically it is applicable for those systems as well, but it wouldn't result in an interesting behavior because it would basically be a kind of trivial thing to apply these frameworks for a simple or simple or simple in the sense of being linear systems.

Thank you, Oli. Very, very deep answer. Michael?

Yeah, I also read in some context that this FEP would explain how a system can come into the steady state, but it cannot explain how it comes out, why, what triggers the system to go out of that steady state.

But this is constantly observed, right? I mean, we are not always in the steady state, but also jump into other states.

So, I don't know whether this is an implication exactly from what Ali said, but I said that's why this free energy principle could not be applied to living systems, even, I think.

Yes, thank you. Well, a lot of perspectives. Let's focus that kind of question on the material of chapter 5, which is the mammalian nervous system and behavior. So, with copious citations and empirical evidence, they're going to be talking about places where the FEP and ACTIM have been applied. And so, this is going to be much more empirical and tangible and anatomical than previous chapters.

So, whereas the generative model proposed in chapter 4, and the high and the low road by which that generative model is reached in chapters 3 and 2 were kind of in principle, here we're going to see a lot more about what it looks like in practice. And that's going to mean that it's in reference to a specific kind of system, like specific brain regions, and it's also going to mean that it's answerable to empirical data, and that it's been published according to professional communication channels, signifying that it had unique explanation or predictive power. So, a lot of these are like fascinating questions about the hard and soft limits of applicability of different methods, and we'll see in chapter 5 how that plays out in practice.

So, again, just write anything or raise your hand if you'd like to explore. So, this chapter is not intended as the final word on process theories. It's simply the interpretation that seems most consistent with the currently available evidence and length constraints on the book.

NORS are aimed to endorse a specific process theory, but to illustrate how the ideas in the previous chapters may be put to work. So, if we were reading an anatomy textbook or a psychology textbook, it would be like, this is how it is. The elbow is connected to this, and this concept does this. But the way that we're going to be looking at the coming examples are representative of how models are constructed that do describe real systems of interest. Yana, I'll bring your question into here and feel free to continue this or add any more information on what type of generalist tools might help. Coda is certainly one generalist tool that helps.

Coda is also one generalist tool that helps. And then there's many statistical approaches that are more technical. So, right now we're going to be approaching this just from the kind of like learning side, but the way that these models are implemented, the code is generally available.

And so those also provide some specific tools. And there may be other tools also, but at least tools that we use for learning help us with Bayesian mechanics in our engagement with it. And then the tools that actually implement those statistical models. And then there may also be other tools as well, which would be interesting to explore. They describe the layout of the chapter, which is going to be first focusing on one of the most well-studied neural systems, which is the mammalian cortex architecture. They're then going to expand into the reflex arc of the spinal cord. And that will help us

look at the continuous time models of perception and action. And then in section 5.4, talking about decision-making.

All right. So in the end of chapter 4, there were some representations that looked like this.

And this is showing two related views on a generative model. On the left is a hierarchical predictive processing model where there are just as a graphical convention, top-down predictions and bottom-up sensory

data. And so this is represented as kind of like a skyscraper, just without reference to time. The tilde means through time. This in practice becomes unrolled through these message passing architectures. And we

talked a little bit about that last week. Here, they're going to return to that idea. So the skyscraper, the kinds of ways in which hierarchical predictive processing architectures can be visualized or represented with a Bayes graph. Those schemes are going to be associated with stereotype circuitry. And there's like

also coming from a long history of taking pictures of the brain and understanding of what the circuitry is in terms of synaptic connectivity and so on. So they're going to describe a little bit about this cortical column. And as an insect neuroscientist, it's fun to read all this because I just think, well, this is one way it could be, but certainly all of these anatomical details, they don't even necessarily have to be that way for mammals, but they definitely don't have to be that way outside of mammals. But connecting it to the reality of what is empirically measured helps ground the theory. And so figure 5.1 is going to be a side-by-side of the anatomy of the cortical layers.

Has this mapping been done for the *C. elegans*? The synaptic connectivity of the, whatever it is, 302 neurons in the *C. elegans*, for the most part has been completed.

So synaptic connectivity maps do exist.

Oh, I mean the sort of the relationship between that map and the message passing that we see here, sort of the mapping between what we're seeing on the left and what we're seeing in the middle and the right. It would be awesome. As far as I know, there's not an explicit

Nematode or even Fruit Fly or any other type of mapping, but I think that's very exciting work.

Because there's like a whole Nematode simulation and there are, with brain and body and biomechanics, and there are similar projects for the *Drosophila* fly. So the bulk of the work of these authors, Tristan et al., and building on the legacy of SPM, most of the work has been done in human neuroanatomy.

And so not many people were considering beyond like the human neuroimaging case until more recently.

So these six layers of the cortex are defined histologically. So just from a tissue definition, they have some morphological differences and they also have like some gene expression and protein expression differences. And these edges represent physical structural connections amongst different regions. It's juxtaposed and these are kind of the outputs, inputs and outputs with these, the thalamus and so on, sending inputs into layer four, and then this kind of resonance processing happening within a columnar unit. And then the outputs of this layer five cell going on to the decision-making areas that are going to be discussed in section five, four. And it's juxtaposed in kind of a visual rhetorical style

with statistical representations about what these populations of cells might be doing.

So that is helping connect the semantics of these variables. Like this one is an error on this, and this one's a variance estimator on that, and this one's getting inputs about this.

All those kinds of things that we want to like talk about what a statistical variable is doing and mapping that onto empirically observed patterns of neuroanatomy.

Now, this is going to be kind of coarse grained. This columnar unit, which is repeated thousands of times in the mammalian cortex, is going to be abstracted from the deep to the superficial.

So this is that layer one, two, three, four, five, six.

Superficial towards the surface of the brain and then the deeper regions of the cortex.

So that is going to be like one unit. In this figure 5.2, they have simplified what's happening within a cortical region to highlight what is happening amongst cortical regions.

And so there's processing happening within a column.

And also columns can serve as bottom up and top down for each other.

So they're interacting laterally from an anatomical perspective.

However, in this representation, the one on the right is like sending top down predictions to the one on the left.

And then bottom up prediction errors or sensory inputs are coming up to the one on the right.

So this is showing that predictive processing architectures can be implemented laterally across cortical columns.

And this is at the root of a lot of current computational modeling efforts.

Just we'll leave it there and we'll return to it.

Can I ask, is that, are they the Taylor series?

Is that what that is, the ascending prediction error?

It's a, I was trying to understand this Taylor series and it seemed to be like a way of creating a set of predictions from a function.

And, you know, so I, I, I, I, whenever I looked at this, I thought that's the Taylor series.

Am I wrong in that?

It's a great insight.

And I think as with, with many of these things, it's like, if you, if you needed and want it to be, you can connect those locations.

So for example, here's figure 4.3.

And in the discrete time case, we were making explicit predictions about the past, present, and future.

In the continuous time case, rather than making explicit predictions about the past, present, and future, we have the x hidden state variable, first derivative, second derivative, and so on, on x .

So that is like the Taylor series coefficients.

Now, one could propose, what if what this cortical column was doing was x ?

So it is getting sensory input.

This is going to be raw sensory input coming in or just from some other location.

And it is going to be doing the role of inference about x .

And then it is going to be constrained and constraining the rate of change in x .

So let's assign the computational function of this middle unit to like the derivative of x .

And this one to the second derivative of x .

So this architecture could implement something like a Taylor series approximation.

You can make it that way in silico, and then you'll know.

Or if you're talking about real brains, of course, they're not actually doing a Taylor series approximation on the coefficients,

but computationally, they might be doing something akin to it.

So.

Thank you for that.

Yeah, thank you.

So, figure 5.1 highlighted how layer 5 of the cortex is going to be projecting out.

Here's layer 5, projecting out to some of these motor and decision-making regions.

So now, we're going to treat the descending prediction from that cortical region.

We're just going to have it up here.

Starting from the BET cells in layer 5 of the motor cortex.

So here's the BET cells.

Here's the motor cortex.

Here's like the brain.

Here's the motor cortex.

Is sending a descending prediction to, in this case, a cross section of the spinal cord.

Sensory data is incoming to the spinal cord through proprioception.

Afferent means away from the sensor.

Efferent means to the sensor.

So, proprioceptive data are coming away from the sensor.

And in the actual histology of the spinal cord,

descending predictions about proprioception, how one is expected to feel,

and then the incoming proprioceptive information are being juxtaposed in this discrepancy variable.

So, subtracting the predictions from the incoming data, so just a simple differencing,

results in an error that says how much muscle contraction would be required to meet the prediction.

So, if the prediction is exactly what the incoming data are, the discrepancy is zero, and no motor action is instigated.

If it's non-zero, motor action would be instigated.

Lower motor neurons then cause this muscle contraction or relaxation, ensuring that the resulting proprioceptive data match descending predictions.

So, we can have continuous.

This is an opportunity and where continuous time models have been studied a lot because the incoming sensory data are continuous.

So, for example, perceived weight or just different tactile sensations.

And then the outgoing behavior can also be framed in a continuous setting, like the elbow angle.

And so, this is like a simple statistical architecture that situates motor action as resolving the discrepancy between predictions incoming and data incoming.

They move on to another region of the mammalian nervous system.

Cortical layer five targets several other structures beyond the spinal cord.

Amongst these is the stratum with these regions.

And a lot is known about that region in terms of gene expression and protein expression and neurophysiology.

So, they're describing some of the empirical neurophysiology, neuroanatomy.

Talking about figure 5.4, it's going to come up several pages later though.

So, just keep that in mind.

They're going to be talking about policy selection in the context of expected free energy.

So, here we didn't need to talk about planning at all.

And continuous time models are very adept at describing these situations where no explicit planning is required.

But still, adaptive action can occur.

And we revisit that in chapter 8 on continuous time models.

But now we're going to look at decision making.

Kind of classical, psychologist-type categorical decision making.

Especially thinking about explicit planning.

And then they're going to talk about how the architecture works.

But we'll talk about that when we see it.

Dopamine tone is going to modulate the balance of the two sides of the figure.

And perturbations of dopamine are going to have some behavioral and cognitive consequences that are completely compatible with this model.

And just in general, the anatomy of figure 5.4 is endorsed by evidence from both clinical pathology and cellular morphology.

So, it's not the only map you can draw of the territory.

It isn't the territory itself.

But the map shown here is compatible with the complexity of the territory.

Okay.

So, now we're going to take those just like they started the section off on.

Outputs of cortical layer 5 are going to target other areas, not just the spinal cord.

So, we're going to have one of those cortical outputs sending into this brain region.

There are going to be two pathways illustrated.

So, one might be in effect or the other.

And dopamine with this gamma variable is modulating the balance between the two.

So, it's kind of like a rudder.

The computational role of dopamine is acting like a rudder with which one of these pathways is being deployed at a given time.

Here are what those two pathways are.

And again, this is going to be like a behavioral computational account.

But it's also consistent with the anatomy and with clinical findings.

What's called the indirect pathway is one where the E vector, which is our habit, the affordances are encoded in E .

So, different opportunities for action are enumerated in E .

And the numbers in each of those cells refer to their prior probability.

So, if there's up, down, left, right, there's going to be four elements in E .

And if it was 1, 1, 1, 1, it means all things are being equal.

There's no prior bias towards the direction to take.

Conversely, in that up, down, left, right case, what if it were 100, 1, 1, 1?

That would be equivalent to saying the prior probability on going up is much higher.

And so, in this indirect pathway, the E affordance vector is basically being passed on through to policy selection.

And so, policy posterior, π , is basically reflecting the policy prior.

So, actions are being selected proportional to how they have been selected in the past.

The distributions of actions being selected in the indirect pathway is being drawn from the same distribution as habit.

So, we can contrast that with what's called by the neurophysiologist the direct pathway.

Here, we see that same kind of input coming in.

But rather than passing E on to π , we have G .

So, G is expected free energy functional.

So, it's equation 2.6 and there's other representations.

And what G is doing is taking in the E , which is like not shown, but it does come into play.

So, this is not exhaustively showing every variable.

It's just trying to only highlight the differences between them, really.

G sharpens the action prior into an action posterior.

G sharpens the action prior.

So, that is updating the policy prior into the posterior through expected free energy.

How does expected free energy sharpen or modify the action prior into the action posterior?

It reweights policies according to their expected free energy.

What is expected free energy?

It's equation 2.6.

So, there's different decompositions.

No one of which is simply like how it is.

But policies are going to be updated.

Becoming to their posterior likelihood from which actions are actually sampled from.

And they're going to be sharpened or updated according to

according to this top line, the pragmatic value and the information gain of different policies.

Michael?

Yeah.

One question.

So, this posterior is still an expected posterior.

And we also have in the variational free energy another kind of posterior.

Is that true?

Every variable in Bayesian statistics that's undergoing updating can be described as having a prior and then data comes in and then there's the posterior.

So, here in particular, we're talking about the prior and the posterior on action policies.

But also, we talk about prior and posterior for other kinds of variables.

Here, the focus is how descending information from the cerebral cortex shapes policy selection.

With the kind of neurophysiological hypothesis, not total expert, that habit is encoded or represented within this brain region.

But cortical inputs shift the balance towards the direct pathway.

So, this is where we see deliberative decision making because we end up sharpening policies according to their expected free energy.

So, let's think about the famous marshmallow example where a marshmallow is placed on the table and someone can be taking it or not.

And they're told about what it is or isn't or anything like that.

So, if somebody had a habit of not taking the marshmallow, then this pathway would lead them to not take it.

If they had a habit of taking it, it would lead them to taking it.

In this case, in the direct pathway, policies would be evaluated according to their epistemic and pragmatic value.

So, in a model where pragmatic value was sugar and there was a preference for sugar, then policy selection would lead to the taking of the marshmallow.

If pragmatic value in that model were being defined as conformity to the rules, then that conformity would be realized by not taking the marshmallow.

So, I know it's kind of hard because we're talking on one hand about actual neurophysiology and behavioral patterns, but on the other hand about these variables, but they're being juxtaposed here to show how it really does come into play in empirical modeling.

Yes, Michael.

But all this is a planning.

This is not based on an observation at that moment, right?

And how does surprise come in there?

So, surprise, how can we also expect surprise and take this into account in this expected free energy?

So, this is showing just a few distilled aspects of this graph.

This is not the full model.

Observations are coming in.

And so, in that sense, surprise being about observations, it could be understood to be happening here.

It's just not being, like, diagrammed.

Yana, what happens when there are more options than a no nutrient stimulant?

Yes, it always...

That's where we get into the figure 9.1.

Not to look ahead, but...

Depending on how the context is set up for the experiment.

Was it a two-choice experiment?

Was it a three-choice experiment?

And then that is the ethologist's challenge.

To develop the as-if view from the inside,

the cognitive model of the subject parameterized by their objective behavior in this setting.

Not the only way to use these models.

Here, they point to how certain dopamine signals that famously, incredibly famously, look like reward prediction errors.

So, one model of dopamine was, like, dopamine is the reward molecule.

And this Schultz 1997 paper was, like, wait a minute.

Yes, when more is received than expected, there's a dopamine spike.

And that was the pattern that people were associating with reward.

But if it's the exact amount that was expected, there's no dopamine spike.

And if it's less than expected, but still a reward, there's, like, another spike.

And so, that was, like, wait a minute.

So, it's not just, like, a thermometer for reward.

Maybe dopamine is doing something like tracking prediction errors.

So, this is kind of like a proto-Bayesian brain, proto-predictive coding, empirical find it.

From the dopamine neurophysiology that we now have, like, much, much more resolution on.

And so, active inference provides a specific alternative explanation to these phenomena.

Where we don't need to make all these exceptions to a reward-based model.

Well, it's dopamine is signaling reward, but then also it does that.

And also, I don't know why it does this.

We can have a unified perspective with active inference.

On empirical data.

Here, they're going to point to the idea that specific hypotheses can be formed.

Another reason to use active inference modeling for these kinds of cognitive, or, I mean, ultimately neurophysiological settings, is that there can be non-invasive measurement of precision parameters. And that's called computational phenotyping. And Ryan Smith and others have done a lot of work on that. Table 5.1 is going to list some of the important neurotransmitters. And then, even though neurotransmitters do different things in different settings, and they're not saying otherwise, they're going to kind of broadly associate it with a given computational function. So, they interact, and there's neuropeptides, and there's all this other stuff happening. But broadly, here are a bunch of lines of evidence that support an understanding of that neurotransmitter associated with precision acting on these things. Thanks for sharing those, Yana. It reminds me of this 2016 paper. They gave ants a choice between sugar water and some dopamine drug. Maybe morphine. Yeah, morphine. And they showed that they could taper down the sugar and the morphine and change a few different things so that there was a persistent choice to go to the morphine. So, I mean, yes. Many, many interesting things to explore. But just to conclude on this part from above, computational phenotyping basically means probing a generative model, reducing our uncertainty about a generative model through peripheral measurements. So, for example, pupil diameter and eye movements contain a lot of information on somebody's arousal and attention. And so, you don't need to have like an implantable neurotech per se. You could use these non-invasive measurements and previously understood relationships like between pupil size and noradrenaline

so that you could actually reduce your uncertainty
about noradrenaline levels
just by looking at the pupil.

They talk about Hebbian plasticity
and different statistical models.

5.6.

Continuous and discrete hierarchies.

This is also
a topic that's
treated a lot more extensively
in Livestream 46.

Active inference models do not contradict folk psychology.

But they highlight
a move from continuous representations
at low levels of a neural hierarchy,
for example, continuous sensory input,
to categorical variables at higher levels,
especially associated with
semantics,
identification,
and decision-making.

So,
we're able to hold categorical beliefs
and interface with
continuously varying
sensory receptors and effectors
like muscle length
or visual features.

That
is just kind of brought up
in these two short paragraphs
in 5.6.

And then
in figure 8.6,
this is an example of what
the generative model looks like.

So,
at the higher level,

there's a missing line here.

But at the higher level,

we have a discrete time generative model,

partially observable

Markov decision process,

POMDP.

How do we know?

Because it has the

$\tau - 1$,

τ ,

$\tau + 1$.

So,

this is making explicit predictions

about the past,

present,

and future.

And in this case,

it's also making

categorical inferences.

So,

is it A or B?

Or,

oh,

it's A and B

because those letters are also used here.

But,

is it raining

or is it sunny?

And we're just going to

have a categorical difference

between those two

and have that estimate

happening past,

present,

and future.

But then,

at each time step,

there's a continuous time model.

And so,
here,
we can recognize
the continuous time model
because we see that
X,
X prime,
X double prime.
and so,
we have like this big rake,
big E pointing down,
and then here's a smaller one.
So,
it's kind of like figure 4.3.
Here's discrete time model,
E pointing down,
continuous time model,
E pointing down,
and then those are going to be
grafted together.
So,
we have a higher order,
slower,
discrete time model,
and then a faster,
continuous time model.
So,
this could reflect,
and this is what they discuss
in Livestream 46
with Alex Kiefer,
Ryan Smith,
and Maxwell Ramstead.
They discuss how
decision making,
like,
should I go to the baker
or to the barber,

is occurring in a discrete-like way
in this decision making model.
But then the implementation
of the decision making model
is happening
in a continuous time model,
which they call
motor active inference,
MAI.

So,
the good thing about
figure 4.3

is
these are
enormously flexible
methods.

This is not like
the one and only way
an active inference model
looks.

It's really
almost
bewilderingly
the opposite of that.

This isn't like
the zoo of what's out there.

These are showing
some of the
structural possibilities
for cognitive modeling.

And there are
many possibilities
and there's many ways
to model
the same
phenomena.

Jana asked,
how is the

continuous
hierarchy
applied?
well,
you could add
more information,
but one could
make a hierarchical
model where
the higher level
was continuous
and the output
of a given
like the Y
here,
we could graph
another continuous
time model
right here.

So,
these are like
our Lego
kits.

These are a few
Lego constructions
that people have
made that
typify
minimal
skeletons
of cognitive
models.

But what's
really important
is that
the cognitive
models that we
build

are going to
have tractable
ways to do
inference on them
because of
message passing.

So,
then they
closed the
chapter with
this summary
visualization.

So,
it's the three
systems that we
learned about
in the chapter,
which is that
cortical
column,
the balance
between the
indirect and
the direct
pathways of
policy selection
in the basal
ganglia
midbrain,
and the
action
selection in
that continuous
time model
happening via
the spinal
cord
discrepancy

detection.

And so,
this is kind
of reflective
and compatible
with the
actual
anatomical
connectivity
of the
body.

So,
it's not
like proposing
a telepathic
link between
some cell
in the brain
and like
the motor
unit in
the arm.

So,
it's compatible
with the
constraints
that exist
in the
body.

It explains
and predicts
different
phenomena
that are
observed.

And those
explanations
come not

from just
trivial,
well,
if two
things that
are touching
are influenced,
then one
influences the
other,
but rather
with actually
these statistical
models that
even help us
understand the
progression or
different kinds
of interventions
in different
conditions,
which is where
these models
have been
used most.

so that's
the end
of that
chapter five.

But they
kind of
close
with
a reminder
that neuroscience
is a work
in progress
and there

are so
many
open
questions
from the
supercellular
level,
like where
are the
glia?
Where's the
immuno
aspect?
To
different
kinds of
everything.
They're just
picking three
examples that
have been
studied in
previous papers
by these
authors and
showing how
alone and
in combination
we can use
these kinds
of models.
So Jana
asked,
so this is
how sensory
experiences can
be modeled.
Broadly,

this could
help us make
sense of
sensory
experience.

This isn't
like the one
and only way
to model
sensory
experience.

this is
like one
circuit board
statistical
model that
can help
with
understanding
how incoming
proprioception
results in
outgoing
action.

So it
can be
used in
service of
understanding
sensory
experience.

But this
is not
the final
model on
sensory
experience.

I hope

that that's

clear.

there can

be other

compositions.

Yes.

So any

other

thoughts or

comments on

this first

pass through

five?

Michael?

Yeah,

but in

this figure

5.5

that you

just

showed,

still the

observation,

the sensation

is actually

missing there,

right?

So we have

action,

but not

the observation.

Here,

the input.

Y is the

proprioceptive

input,

but other

kinds like

visual input
is not
shown here.
but that
speaks to
the
composability
of these
models.
We could
have an
auditory
stream and
a visual
stream and
a proprioceptive
stream,
and we can
do model
comparison and
we can
develop models
that use
any mixtures
of those,
and on
given
computational
hardware for
a given
data set,
for a given
setting,
for a given
client,
one of those
models might
be more

useful than

the other

for you.

Okay,

thank you.

Yeah,

it's a fun

chapter,

and it's

definitely a

context switch

because chapter

four just

goes so hard

with the

technical

details,

and then

5.3 is

really about

our bodies,

but in

that contrast,

there's been a

lot of great

discussion.

So,

next week,

we'll return

for the second

discussion on

5.

Please feel free

to ask your

questions in

the meanwhile.

So,

thank you,

everybody.

See you next
time.